AMRITA SAI
INSTITUTE OF SCIENCE & TECHNOLOGY

SKILLATHON

PROJECT DOCUMENTATION

Project Title: HDFS Commands

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Team Number: 2

Team Members:Bandi Naga Manju Bhargavi(23aj1a0509)
Nemalidinne Tharangini (23aj1a0583)

Introduction: HDFS (Hadoop Distributed File System) is a distributed storage system used to store large amounts of data across multiple machines. It is fault-tolerant and scalable. It stores data in multiple nodes and ensures reliability.

Architecture Diagram:

```
              NameNode
                 |
      -------------------------
      |          |           |
  DataNode   DataNode    DataNode
```

NameNode manages metadata, while DataNodes store actual data blocks.

Tools / Components Used: Hadoop, Java, Linux, Command Line Interface, Sample Files

Explanation:

1. File System Navigation: It is used to move through directories and manage folders in HDFS. In mobile phone, opening file manager and navigating between folders is similar. Commands like -ls, -mkdir help to list and create directories. Example: Creating a new folder in phone is similar to creating directory in HDFS.

2. File Operations: It includes uploading, downloading, and viewing files. In mobile phone, uploading photos to cloud or downloading files is similar. Commands like -put, -get, -cat are used. Example: Uploading a file is like sending photo to Google Drive.

3. File Management: It includes copying, moving, and deleting files. In mobile phone, copying or deleting files from storage is similar. Commands like -cp, -mv, -rm are used. Example: Moving a file from one folder to another in phone.

4. Disk Usage: It is used to check storage information like file size and available space. In mobile phone, checking storage usage in settings is similar. Commands like -du, -df are used. Example: Checking how much storage is used in phone.

5. Permissions: It is used to control access to files. In mobile phone, giving app permissions or locking apps is similar. Commands like -chmod, -chown are used. Example: Restricting access to files for security.

2. File System Navigation

• hdfs dfs -ls /path : Lists files and directories in the given path.
Example: hdfs dfs -ls /user/data

• hdfs dfs -ls -R /path : Lists all files recursively.

• hdfs dfs -mkdir /path : Creates a directory.
Example: hdfs dfs -mkdir /user/newfolder

• hdfs dfs -mkdir -p /path/subdir : Creates nested directories.

• hdfs dfs -rmdir /path : Removes an empty directory.

3. File Operations

• hdfs dfs -put localfile /hdfs/path : Uploads file from local system to HDFS.
Example: hdfs dfs -put data.txt /user/data

• hdfs dfs -get /hdfs/file localpath : Downloads file from HDFS to local system.

• hdfs dfs -cat /file : Displays file content.

• hdfs dfs -head /file : Shows first few lines of file.

• hdfs dfs -tail /file : Shows last few lines of file.

4. File Management

• hdfs dfs -cp /source /destination : Copies file within HDFS.

• hdfs dfs -mv /source /destination : Moves file.

• hdfs dfs -rm /file : Deletes file.

• hdfs dfs -rm -skipTrash /file : Permanently deletes file.

5. Disk Usage & Information

• hdfs dfs -du /path : Shows file size.

• hdfs dfs -df -h : Displays disk usage in human-readable format.

• hdfs dfs -count /path : Shows number of files, directories, and bytes.

6. Permissions

• hdfs dfs -chmod 777 /path : Changes permissions.

• hdfs dfs -chown user:group /path : Changes owner.

• hdfs dfs -chgrp group /path : Changes group.

7. Conclusion

HDFS commands help in managing files and directories efficiently in a distributed environment. Understanding these basic commands is essential for working with Hadoop systems.